# Quarry: Technical Architecture Specification

Local Semantic Search for AI Agents

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# 1 Overview

Quarry is a local semantic search engine. It ingests documents in 20+ formats, embeds them with a local ONNX model, stores vectors in LanceDB, and serves semantic search to AI agents (Claude Code, Claude Desktop) and a macOS menu bar app.

The system follows the Punt Labs projection strategy: a single core library projects outward through four thin adapter layers. Each surface imports core functions and calls them with resolved arguments. No intermediary service layer.

## 1.1 Design Principles

**P1**. **Library-first.** The core library does all work. CLI, MCP, HTTP, and plugin surfaces are thin wrappers that resolve settings and delegate.

**P2**. **Local by default.** Everything runs on the user's machine. No API keys, no cloud accounts. The embedding model downloads once (∼120 MB, int8 quantized).

**P3**. **One daemon, many clients.** A single `quarry serve` process loads the embedding model once and serves all sessions over WebSocket. Clients connect via `mcp-proxy`.

**P4**. **Fire-and-forget I/O.** Side-effect MCP tools return immediately and process in a bounded thread pool. Search is synchronous.

**P5**. **Named databases.** Each database is fully isolated: its own LanceDB directory, sync registry, and vector index. Work/personal separation without configuration files.

# 2 System Architecture

## 2.1 Deployment Topology

```
                  stdio WebSocket
Claude Code <------------------> mcp-proxy <-------------------> quarry serve
            MCP JSON-RPC (~5 MB Go) (one daemon)

Claude Desktop <--- stdio ---> quarry mcp
                              (standalone process, no daemon)
```

Without the proxy, every Claude Code tab spawns a separate Python process, each loading the embedding model into ∼200 MB of RAM. With it, startup is instant and state is shared across all sessions.

The CLI (`quarry find`, `quarry ingest`, etc.) uses the library directly — no daemon involved. Each invocation loads the embedding model, does the work, and exits.

## 2.2 Surface Summary

| Surface | Transport | Purpose |
|---|---|---|
| CLI | Direct library calls | Interactive use from terminal |
| MCP server | stdio or WebSocket | Claude Code/Desktop tools |
| HTTP server | REST JSON + WS `/mcp` | Menu bar app + mcp-proxy |
| Plugin | Hook shell commands | Auto-register, auto-ingest, compaction safety |

# 3 Module Responsibilities

## 3.1 Data Layer

| Module | File | Responsibility |
|---|---|---|
| Models | `src/quarry/models.py` | Immutable dataclasses: `PageType` enum, `PageContent`, `Chunk`, `PageAnalysis`. All `@dataclass(frozen=True)`. |
| Types | `src/quarry/types.py` | Protocol definitions for external dependencies: `LanceDB`, `LanceTable`, `LanceQuery`, `ListTablesResult`, `OcrBackend`, `EmbeddingBackend`. Structural typing instead of `Any`. |
| Config | `src/quarry/config.py` | `Settings` (pydantic `BaseSettings`), `resolve_db_paths()` for named databases, ONNX model constants. |
| Database | `src/quarry/database.py` | LanceDB operations: `insert_chunks`, `search`, `list_documents`, `delete_document`, `count_chunks`, etc. Single table `chunks`. |

## 3.2 Ingestion Pipeline

| Module | File | Responsibility |
|---|---|---|
| Pipeline | `src/quarry/pipeline.py` | Orchestrator: dispatches by format, chunks, embeds, stores. Entry points: `ingest_document`, `ingest_content`, `ingest_auto` (URL with sitemap discovery). |
| Chunker | `src/quarry/chunker.py` | Text splitting with overlap. Respects page boundaries and heading structure. |
| Collections | `src/quarry/collections.py` | `derive_collection()` — determines collection name from path or URL. |

## 3.3 Format Processors

Each processor converts a source format into `list[PageContent]`:

| Module | File | Formats |
|---|---|---|
| PDF | `src/quarry/pdf_analyzer.py` | PDF (text via PyMuPDF, OCR fallback for image pages) |
| Image | `src/quarry/image_analyzer.py` | PNG, JPG, TIFF, BMP, WebP (OCR) |
| Text | `src/quarry/text_processor.py` | TXT, MD, LaTeX, DOCX |
| Code | `src/quarry/code_processor.py` | 30+ languages (AST-aware splitting) |
| HTML | `src/quarry/html_processor.py` | HTML (boilerplate stripping, Markdown conversion) |
| Spreadsheet | `src/quarry/spreadsheet_processor.py` | XLSX, CSV (tabular serialization) |
| Presentation | `src/quarry/presentation_processor.py` | PPTX (slide-per-chunk with notes) |
| Dispatcher | `src/quarry/text_extractor.py` | Extracts text from text-based PDF pages (PyMuPDF) |

### 3.4 Backends

| Module | File | Responsibility |
|---|---|---|
| Factory | `src/quarry/backends.py` | Thread-safe factory with double-checked locking. `get_ocr_backend()` and `get_embedding_backend()` return cached singletons. |
| OCR | `src/quarry/ocr_local.py` | Local OCR via RapidOCR (offline, no setup). |
| Embeddings | `src/quarry/embeddings.py` | Local ONNX embedding (snowflake-arctic-embed-m-v1.5, 768-dim, 512 tokens). |

### 3.5 Sync and Registry

| Module | File | Responsibility |
|---|---|---|
| Sync | `src/quarry/sync.py` | Directory sync: discover files, compute delta (new/changed/deleted), ingest/delete accordingly. `ThreadPoolExecutor` for concurrent ingestion. |
| Registry | `src/quarry/sync_registry.py` | SQLite registry of watched directories. Maps directories to collections with timestamps. |

### 3.6 Surfaces

| Module | File | Responsibility |
|---|---|---|
| CLI | `src/quarry/__main__.py` | Typer CLI with rich formatting. Synchronous. All commands: resolve settings → call core → format output. |
| MCP Server | `src/quarry/mcp_server.py` | FastMCP server. Read-only tools are synchronous. Side-effect tools use fire-and-forget. |
| HTTP Server | `src/quarry/http_server.py` | Starlette ASGI app on uvicorn. REST endpoints + WebSocket `/mcp`. |
| Hooks | `src/quarry/hooks.py` | Claude Code plugin hooks: session-start (auto-register), post-web-fetch (auto-ingest), pre-compact (capture transcript). All fail-open. |

### 3.7 Support

| Module | File | Responsibility |
|---|---|---|
| Formatting | `src/quarry/formatting.py` | Pre-formatted plain text output using unicode box-drawing. Used by both CLI and MCP for consistent output. |
| Results | `src/quarry/results.py` | `SyncResult` dataclass for sync operation reporting. |
| Sitemap | `src/quarry/sitemap.py` | Sitemap discovery and URL extraction for web ingestion. |
| Doctor | `src/quarry/doctor.py` | Health checks: model availability, database access, backend connectivity. |

## 4 Daemon Model

### 4.1 Why a Daemon

The embedding model (snowflake-arctic-embed-m-v1.5, int8 quantized ONNX) occupies ∼200 MB of RAM and takes ∼250 ms to load. MCP's stdio transport model spawns a new process per session. Without a daemon, five Claude Code tabs means five copies of the model in memory and five cold starts.

`quarry serve` loads the model once and keeps it resident. The `mcp-proxy` binary (∼5 MB Go binary, <10 ms startup) bridges each session's MCP stdio to the daemon over WebSocket.

### 4.2 mcp-proxy Architecture

`quarry install` downloads `mcp-proxy` (SHA256-verified, platform-specific binary) and configures MCP clients to use it. The proxy:

1. Reads MCP JSON-RPC from stdin.
2. Forwards to `ws://localhost:8420/mcp` over WebSocket.
3. Relays responses back to stdout.

Each WebSocket connection gets its own asyncio Task with a `ContextVar` for database selection (`_db_name`), so `use("work")` in one session does not affect others.

### 4.3 CLI Independence

The CLI does not use the daemon. `quarry find "margins"` loads the embedding model in-process, queries LanceDB, prints results, and exits. LanceDB handles concurrent reads safely, so CLI and daemon can access the same database simultaneously.

## 5 MCP Tool Surface

| Tool | Key Parameters | Purpose | Exec |
|---|---|---|---|
| `find` | `query`, `limit`, `collection` | Semantic search | Sync |
| `ingest` | `source`, `overwrite`, `collection` | Ingest file or URL | Bg |
| `remember` | `content`, `document_name` | Ingest inline text | Bg |
| `list` | `kind` | List indexed data | Sync |
| `show` | `document_name`, `page_number` | Document metadata or page | Sync |
| `delete` | `name`, `kind` | Delete document or collection | Bg |
| `register_directory` | `directory`, `collection` | Register for sync | Bg |
| `deregister_directory` | `collection`, `keep_data` | Remove registration | Bg |
| `sync_all` | — | Sync all registered dirs | Bg |
| `status` | — | Database stats | Sync |
| `use` | `name` | Switch active database | Sync |

Bg = background (`ThreadPoolExecutor`).

Background tools use a bounded `ThreadPoolExecutor(max_workers=4)`. They return an optimistic response immediately. Settings and database connections are snapshotted at the tool boundary and passed into background functions — this avoids a race condition where the mutable `_db_name` ContextVar could change between the tool call and the background thread's execution. Exceptions in background threads are logged via `logger.exception()`, not raised.

## 6 HTTP Server

### 6.1 ASGI Application

Starlette app built by `build_app()` factory, served by uvicorn. REST handlers are sync functions — Starlette auto-detects this and runs them in its threadpool. The `/mcp` WebSocket endpoint is async.

Default port: 8420 (`DEFAULT_PORT` in `src/quarry/config.py`), overridable with `--port`.

## 6.2 Endpoints

| Path | Method | Purpose | Auth |
|---|---|---|---|
| /health | GET | {"status":"ok", "uptime_seconds":N} | No |
| /search | GET | Semantic search (q, limit, collection) | Yes |
| /documents | GET | List indexed documents | Yes |
| /collections | GET | List collections | Yes |
| /status | GET | Database statistics | Yes |
| /mcp | WS | MCP JSON-RPC over WebSocket | Yes |

## 6.3 Port File

After binding, the server writes the actual port to ~/.quarry/data/<db>/serve.port. This supports --port 0 (ephemeral assignment). Removed on shutdown via lifespan context.

# 7 Configuration

## 7.1 Settings

All settings are environment variables read via pydantic-settings.

| Variable | Default | Description |
|---|---|---|
| **Paths** | | |
| QUARRY_ROOT | ~/.quarry/data | Base directory for all databases |
| LANCEDB_PATH | <root>/default/lancedb | Vector database location |
| REGISTRY_PATH | <root>/default/registry.db | Directory sync SQLite database |
| LOG_PATH | ~/.quarry/data/quarry.log | Log file (rotating, 5 MB, 3 backups; independent of QUARRY_ROOT) |
| **Chunking** | | |
| CHUNK_MAX_CHARS | 1800 | Target max characters per chunk (∼450 tokens) |
| CHUNK_OVERLAP_CHARS | 200 | Overlap between consecutive chunks |
| **Embedding** | | |
| EMBEDDING_MODEL | Snowflake/snowflake-arctic-embed-m-v1.5 | Model identifier (display only — the ONNX model is fixed) |
| EMBEDDING_DIMENSION | 768 | Vector dimension (display only — fixed at 768) |

## 7.2 Named Databases

resolve_db_paths(settings, name) maps a database name to a path under QUARRY_ROOT. Each database is fully isolated — its own LanceDB directory, sync registry, and vector index. Names containing path separators (/ or \) or the literal traversal segments . or .. are rejected; names like foo.bar are allowed. If LANCEDB_PATH was explicitly overridden, the override is preserved.

The persistent default database is stored in ~/.quarry/config.toml as [default] database = "name".

## 7.3 Chunking Tuning

**Increase** CHUNK_MAX_CHARS (3000–4000) for documents with long self-contained sections: legal contracts, academic papers, large source code functions.

**Decrease** CHUNK_MAX_CHARS (800–1200) for dense documents where each paragraph covers a distinct topic.

**Increase** `CHUNK_OVERLAP_CHARS` (400) when relevant information spans paragraph boundaries.

The embedding model has a 512-token context window. Chunks exceeding this are truncated during embedding, reducing quality for the tail.

# 8 Search and Retrieval

## 8.1 Embedding Model

snowflake-arctic-embed-m-v1.5 via ONNX Runtime:

- 768-dimensional vectors, 512-token context window
- Asymmetric retrieval: queries are prefixed with a search instruction, documents are not
- int8 quantized ONNX (∼120 MB on disk)
- Runs on CPU only (∼50 ms per chunk on Apple Silicon)
- Auto-downloads on first use via `huggingface_hub`

The model is fixed — there is no configuration to swap it. Changing the model invalidates all existing embeddings, requiring full re-ingestion.

## 8.2 OCR

RapidOCR handles all image-based text extraction. For semantic search, OCR accuracy matters less than expected — the embedding model is robust to minor OCR errors. A misspelled word rarely changes semantic meaning enough to affect search ranking.

## 8.3 Search Quality Tips

1. **Use natural language queries.** The model is trained on question-passage pairs. "What were Q3 revenue figures?" works better than "Q3 revenue."
2. **Use collection filtering.** `quarry find "auth logic" -c backend`.
3. **Re-ingest after tuning.** Chunk parameters only affect new ingestions. Run `quarry sync` after changes.
4. **Check chunk boundaries.** `quarry find -n 1 "query"` and examine the result.
5. **Use `--overwrite` when re-ingesting.** Without it, duplicate chunks accumulate.

# 9 Logging and Exceptions

## 9.1 Logging Standard

Every module that performs I/O declares `logger = logging.getLogger(__name__)`. Pure data modules (`models.py`, `types.py`, `chunker.py`) do not.

Logging is configured once at the application entry point (CLI or MCP server). Library modules never call `logging.basicConfig`.

| Level | Use |
|---|---|
| `DEBUG` | Internal state: variable values, branch decisions, intermediate results. |
| `INFO` | Operational milestones: starting a job, completing a step, resource counts. |
| `WARNING` | Unexpected but recoverable: empty input, deprecated parameter usage. |
| `ERROR` | Failures that prevent completion but do not crash. Always with exception context. |

`CRITICAL` is not used. If the process must exit, it raises an exception.

Log messages use `%s`-style formatting (lazy evaluation), not f-strings. Each message includes enough context to trace an operation without reading code.

### 9.2 Exception Handling

The codebase uses built-in exception types. No custom exceptions unless a caller needs to distinguish failure modes programmatically.

| Exception | Raised when |
|---|---|
| `ValueError` | Invalid input: unsupported format, bad parameter. |
| `FileNotFoundError` | A file path argument does not exist. |
| `RuntimeError` | An operation reports failure (OCR engine error, ingestion failure). |
| `TimeoutError` | An external service exceeds its time budget. |

**Rules:**

1. Never swallow exceptions. Every `except` block must re-raise, or log with `logger.exception()` and raise a different exception, or (at system boundaries only) return a documented sentinel.
2. Never use bare `except:`.
3. Never catch `Exception` broadly in library code. Broad catches are permitted only at the outermost boundary (MCP tool handler, CLI command).
4. Exceptions carry context: include the values that caused the failure.
5. Use `try/finally` for cleanup, not `try/except`.
6. Do not use exceptions for flow control.

## 10 Security

### 10.1 Bearer Token Authentication

`quarry serve --api-key` (or `QUARRY_API_KEY` env var) gates all HTTP endpoints except `/health` behind `Authorization: Bearer <key>`.

- `/health` is exempt — load balancers need unauthenticated health checks.
- `OPTIONS` is exempt — CORS preflight requests carry no auth.
- `hmac.compare_digest` for token comparison — prevents timing attacks.
- Case-insensitive scheme per RFC 7235.
- Empty key = no auth — prevents accidental trivial bypass.
- No key = auth disabled — local use requires no key.

The server refuses to bind to a non-loopback address without `--api-key`.

### 10.2 WebSocket Security (CSWSH)

Origin-based cross-site WebSocket hijacking protection:

- If an `Origin` header is present, it must match the CORS allow-list.
- Browsers always send `Origin`; non-browser clients like `mcp-proxy` do not.
- Bearer auth is checked before WebSocket accept (close code 1008 on failure).

### 10.3 CORS

`CORSMiddleware` configured with GET and OPTIONS methods, Authorization and Content-Type headers. Default allowed origin: `http://localhost`. Configurable via `--cors-origin` (repeatable).

The server reflects the request's `Origin` header only when it matches the allow list, and adds `Vary: Origin` to signal cacheability varies by origin. When no `Origin` is present, no CORS headers are emitted.

### 10.4 Log Safety (CWE-532, CWE-117)

Uvicorn's access log is disabled (`access_log=False`), eliminating query string leakage at the source. The search handler logs only result count, never the raw query. WebSocket session keys are sanitized before logging: control characters stripped via `_CONTROL_CHAR_RE` and truncated to 64 characters to prevent log injection (CWE-117) and log flooding.

## 11 Deployment

### 11.1 Local (CLI)

```
uv tool install punt-quarry
quarry install # downloads model + mcp-proxy
quarry doctor # verifies environment
```

### 11.2 Daemon (`quarry serve`)

```
quarry serve --port 8420 --api-key $QUARRY_API_KEY
```

Managed by `launchd` (macOS) or `systemd` (Linux) via `quarry install`.

### 11.3 Container (Fly.io)

Multi-stage Dockerfile, three stages:

1. **deps** (`python:3.13-slim`): installs `uv`, runs `uv sync --frozen --no-dev`. Two-pass for Docker layer caching (deps first, then source).
2. **model** (extends `deps`): downloads the embedding model at build time (∼120 MB, int8 quantized). Cached as a separate Docker layer.
3. **runtime** (`python:3.13-slim`): copies `/app` from deps, copies model cache from model stage. `QUARRY_ROOT=/data`, port 8080.

LanceDB data lives on a Fly persistent volume at `/data`. `QUARRY_API_KEY` is set via `fly secrets`. Cold start is ∼5 s with the baked-in model.

## 12 Performance

### 12.1 Critical Path

| Operation | Latency | Notes |
|---|---|---|
| Model load | ∼250 ms | One-time per process |
| Embedding (1 chunk) | ∼50 ms | Apple Silicon, CPU |
| LanceDB search | <10 ms | Vector similarity + filter |
| End-to-end search | <100 ms | After model is loaded |

### 12.2 Fire-and-Forget MCP Pattern

Side-effect MCP tools return immediately and process in a bounded thread pool (4 workers). This is necessary because:

1. MCP tool calls block the LLM's response stream. A 30-second ingest blocks Claude from responding.
2. The bounded pool prevents resource exhaustion under concurrent requests.
3. Settings and database connections are snapshotted at the tool boundary, avoiding race conditions with the mutable `_db_name` ContextVar.

### 12.3 Single Table Design

All chunks live in one LanceDB table (`chunks`). Document and collection boundaries are columns, not separate tables. This simplifies cross-document search and avoids table proliferation. Filtering by document/collection/page_type/source_format uses LanceDB's built-in filter predicates on the vector search.

## 13 Plugin and Hook System

The Claude Code plugin (`.claude-plugin/`) registers hooks that fire on session events and tool calls. All hooks are fail-open: exceptions are logged and the process exits 0.

| Hook Event | Matcher | Effect | Blocks? |
|---|---|---|---|
| SessionStart | * | Auto-register directory + background sync | No |
| PostToolUse | quarry tools | Suppress tool output formatting | No |
| PostToolUse | WebFetch | Auto-ingest fetched URL | No |
| PreCompact | * | Capture transcript as session notes | No |

### 13.1 Session Start

`handle_session_start()` auto-registers the working directory and launches a background sync via a detached subprocess. Uses atomic PID-file locking (`O_CREAT|O_EXCL`) at `~/.quarry/sync.pid` to prevent concurrent syncs. Returns `additionalContext` immediately without waiting.

### 13.2 Web Fetch Capture

`handle_post_web_fetch()` ingests URLs fetched during a session into the `web-captures` collection. Deduplicates by document name. Prefers already-fetched content from the tool response (avoids re-fetch, reduces SSRF surface).

### 13.3 Pre-Compact Capture

`handle_pre_compact()` reads the transcript (JSONL) and ingests it into the `session-notes` collection. Extracts user/assistant message text, skipping tool-use blocks. Capped at 500,000 characters.

## 14 CLI Commands

| Command | Purpose |
|---|---|
| `ingest <source>` | Ingest file or URL |
| `remember` | Ingest stdin content |
| `find <query>` | Semantic search |
| `show <document>` | Document metadata or page text |

| Command | Purpose |
|---|---|
| `status` | Database statistics |
| `use <name>` | Set persistent default database |
| `delete <name>` | Delete document or collection |
| `register <directory>` | Register for incremental sync |
| `deregister <collection>` | Remove directory registration |
| `sync` | Sync all registered directories |
| `list documents` | List indexed documents |
| `list collections` | List collections with counts |
| `list databases` | List named databases |
| `list registrations` | List registered directories |
| `serve` | Start HTTP API server |
| `mcp` | Start MCP server (stdio) |
| `install` | Set up data directory, download model |
| `doctor` | Check environment health |
| `uninstall` | Remove system daemon |
| `version` | Print version |

Global flags: `--json`, `--verbose/-v`, `--quiet/-q`, `--db <name>`.

# 15 Test Architecture

821 tests across 33 files. Three tiers plus integration:

| Tier | Strategy | Files |
|---|---|---|
| Tier 1: Core logic | Real data, no mocks | `test_database`, `test_chunker`, `test_embeddings`, `test_code_processor`, `test_html_processor`, `test_text_processor`, `test_collections`, `test_config`, `test_formatting`, `test_image_analyzer`, `test_latex_utils`, `test_models`, `test_pdf_analyzer`, `test_presentation_processor`, `test_spreadsheet_processor`, `test_text_extractor`, `test_registry` (17 files) |
| Tier 2: Orchestration | Mocked I/O | `test_pipeline`, `test_pipeline_images`, `test_ocr_local`, `test_url_ingestion`, `test_sitemap`, `test_backends`, `test_sync`, `test_hooks`, `test_doctor` (9 files) |
| Tier 3: Surface wiring | Fully mocked, <2 s | `test_cli`, `test_mcp_server`, `test_mcp_websocket`, `test_http_server`, `test_proxy`, `test_service` (6 files) |
| Integration | Real ONNX + real LanceDB | `test_integration` (`@pytest.mark.slow`) |

Quality gates: `ruff` (lint + format), `mypy` (strict), `pyright` (strict), `pytest` (all markers except slow/integration). Zero violations required.

## 16 Formal Model

The plugin state machine — how quarry hooks into the Claude Code session lifecycle — is formally specified in `docs/claude-code-quarry.tex` using Z notation (ISO 13568).

The specification covers:

- Session start with/without auto-sync
- Web fetch auto-capture into `web-captures` collection
- Pre-compaction transcript capture into `session-notes`
- Database switching via `use` tool
- Knowledge hint injection at tool boundaries
- Compaction observability: a capture flag records whether capture preceded compaction; the hook system enforces capture-before-compaction on a best-effort basis
- Clean session boundaries: all quarry state cleared on `EndSession`

The specification can be type-checked with `fuzz -t claude-code-quarry.tex`.